

Local Fréchet regression with spherical predictors

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August 20, 2026

1 The problem

Many nonparametric regression methods are designed for Euclidean responses and predictors. Fréchet regression allows the response to take values in a general metric space, but most existing methods use Euclidean predictors. This paper studies Fréchet regression with a spherical predictor

$$X \in \mathbb{S}^d = \left\{ u \in \mathbb{R}^{d+1} : \|u\|_2 = 1 \right\}$$

and a response

$$Y \in \mathbb{M},$$

where $(\mathbb{M}, d_{\mathbb{M}})$ is a metric space.

2 The setup

When $\mathbb{M} = \mathbb{R}$, the usual regression function can be expressed as

$$E(Y \mid X = x) = \arg \min_{y \in \mathbb{R}} E \left\{ (Y - y)^2 \mid X = x \right\}.$$

For a general metric-space-valued response, define

$$M_{\oplus}(x, y) = E \left\{ d_{\mathbb{M}}^2(Y, y) \mid X = x \right\}.$$

The target regression function is

$$m_{\oplus}(x) = \arg \min_{y \in \mathbb{M}} M_{\oplus}(x, y).$$

Given i.i.d. observations

$$\{(X_i, Y_i) : 1 \leq i \leq n\},$$

the goal is to estimate $m_{\oplus}(x)$ for $x \in \mathbb{S}^d$.

3 The main idea

The paper constructs local Fréchet regression estimators using the geometry of the sphere.

For the local constant estimator, observations are weighted according to their spherical closeness to x . Since the angle between X_i and x is

$$\arccos(X_i^{\top} x),$$

the quantity

$$1 - X_i^{\top} x$$

is small when X_i is close to x . For a kernel L and bandwidth h , the local constant objective function is

$$\widehat{M}_{h,0}(x, y) = \frac{n^{-1} \sum_{i=1}^n L\left(\frac{1 - X_i^\top x}{h^2}\right) d_{\mathbb{M}}^2(Y_i, y)}{n^{-1} \sum_{i=1}^n L\left(\frac{1 - X_i^\top x}{h^2}\right)}.$$

The local constant Fréchet estimator is

$$\widehat{m}_{h,0}(x) = \arg \min_{y \in \mathbb{M}} \widehat{M}_{h,0}(x, y).$$

The paper also proposes a local linear Fréchet estimator. A tangent-normal decomposition around $x \in \mathbb{S}^d$ gives a first-order approximation for nearby points and motivates spherical local linear weights. The objective function is

$$\widehat{M}_{h,1}(x, y) = n^{-1} \sum_{i=1}^n \widehat{W}_{x,h}(X_i) d_{\mathbb{M}}^2(Y_i, y),$$

where $\widehat{W}_{x,h}(X_i)$ is the spherical local linear weight. The estimator is

$$\widehat{m}_{h,1}(x) = \arg \min_{y \in \mathbb{M}} \widehat{M}_{h,1}(x, y).$$

Both estimators minimize weighted Fréchet objective functions. The local linear estimator additionally uses tangent-space information to obtain a first-order local approximation.

4 The main results

The paper establishes consistency and convergence rates for both $\widehat{m}_{h,0}(x)$ and $\widehat{m}_{h,1}(x)$. Simulation studies consider spherical-circular and spherical-spherical regression, and the local linear estimator performs best in the reported settings. The method is also applied to distributional-circular, distributional-spherical, and network-circular regression.

5 My takeaways

- Why are the weights based on $1 - X_i^\top x$ and scaled by h^2 , rather than being written directly in terms of the geodesic distance $\arccos(X_i^\top x)$?
- What conditions on the metric space $\mathbb{M}$ ensure that the Fréchet minimizers defining $m_{\oplus}(x)$, $\widehat{m}_{h,0}(x)$, and $\widehat{m}_{h,1}(x)$ are unique?

References

- [1] C. J. Im, J. M. Jeon, and B. U. Park. Local Fréchet regression with spherical predictors. *Electronic Journal of Statistics*, 19(2):5313–5367, 2025. <https://doi.org/10.1214/25-EJS2462>.